

Hall Ticket Number:

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Code No. : 13258 N/O

**VASAVI COLLEGE OF ENGINEERING (AUTONOMOUS), HYDERABAD***Accredited by NAAC with A++ Grade***B.E. III-Semester Main & Backlog Examinations, February/March-2023****Computer Architecture**

(Common to CSE &amp; AIML)

Time: 3 hours

Max. Marks: 60

*Note: Answer all questions from Part-A and any FIVE from Part-B***Part-A (10 × 2 = 20 Marks)**

| Q. No.                           | Stem of the question                                                                                                                                                                                                                                                                                                                                               | M | L | CO | PO |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|----|----|
| 1.                               | Compare and contrast computer architecture with computer organization?                                                                                                                                                                                                                                                                                             | 2 | 2 | 1  | 1  |
| 2.                               | Draw and explain the block diagram and timing diagram for the statement.<br>P1: R2 ← R1                                                                                                                                                                                                                                                                            | 2 | 1 | 1  | 2  |
| 3.                               | List out the advantages and disadvantages of hard-wired and micro-programmed control unit?                                                                                                                                                                                                                                                                         | 2 | 1 | 2  | 1  |
| 4.                               | Explain the following RTL concerned to instruction BSA?<br>D <sub>5</sub> T <sub>4</sub> : M[AR] ← PC, AR ← AR+1<br>D <sub>5</sub> T <sub>5</sub> : PC ← AR, SC ← 0                                                                                                                                                                                                | 2 | 2 | 2  | 2  |
| 5.                               | What is parallel processing? Explain Flynn's classification of computer?                                                                                                                                                                                                                                                                                           | 2 | 1 | 3  | 1  |
| 6.                               | A non-pipeline system takes 100ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 20ns. Determine the speedup ratio of the pipeline for 200 tasks?                                                                                                                                                               | 2 | 2 | 3  | 2  |
| 7.                               | What are the values of input ready, output ready, and control bits F1 through F4, when (a) The buffer is empty? (b) The buffer is full?                                                                                                                                                                                                                            | 2 | 1 | 4  | 1  |
| 8.                               | Define bit-oriented protocol? Determine the minimum and maximum number of bits that a frame must have in the bit-oriented protocol?                                                                                                                                                                                                                                | 2 | 2 | 4  | 2  |
| 9.                               | Determine an average access time of a computer with cache access time of 100 ns, a main memory access time of 1000 ns, and a hit ratio of 0.9?                                                                                                                                                                                                                     | 2 | 2 | 5  | 2  |
| 10.                              | State Amdahl's Law? If a system has a single bottleneck that occupies 25% of the total execution time, and we add 5 more processors to the system, find the speedup?                                                                                                                                                                                               | 2 | 2 | 5  | 2  |
| <b>Part-B (5 × 8 = 40 Marks)</b> |                                                                                                                                                                                                                                                                                                                                                                    |   |   |    |    |
| 11. a)                           | Design 4-bits adder/subtractor using full adder with suitable gates?                                                                                                                                                                                                                                                                                               | 4 | 3 | 1  | 3  |
| b)                               | Let the contents of 8-bit registers A, B, C, and D initially are 11110010, 11111111, 10111001, and 11101010 respectively. Determine the 8-bit values in each register after performing following sequence of microoperations:<br>A ← A + B // Add B to A<br>C ← C ^ D // AND D to C<br>B ← B + 1 // Increment B<br>A ← A - C // Subtract C from A                  | 4 | 2 | 1  | 2  |
| 12. a)                           | Explain control memory and the associated hardware needed for selecting the next microinstruction address with neat block diagram?                                                                                                                                                                                                                                 | 4 | 1 | 2  | 1  |
| b)                               | A Computer uses a memory unit with 256K word of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts: an indirect bit, an operation code, and a register code part to specify one of 64 registers and an address part. How many bits are there in operation code, the register code part, and the address part? | 4 | 3 | 2  | 3  |

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|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|-----|
| 13. a) | What are the addressing modes? Explain different types of addressing modes with suitable examples?                                                                                                                                                                                                                                                                                                                                                                                                                    | 4 | 1 | 3 | 1   |
| b)     | Draw the block diagram of an 8-bit ALU with a 4-bit status register? Explain four status bits (C, S, Z, and V) in brief? Determine the status of 4-bits register after execution of following instructions. Assume that the value of 8-bit register R1 in each case is 9C (Hexadecimal)?<br>(a) Add immediate operand 1B to R1<br>(b) Subtract immediate operand 7A from R1                                                                                                                                           | 4 | 3 | 3 | 3   |
| 14. a) | Explain Asynchronous Communication Interface or Universal Asynchronous Receiver-transmitter (UART) with neat block diagram and function table?                                                                                                                                                                                                                                                                                                                                                                        | 4 | 2 | 4 | 2   |
| b)     | Obtain the truth table of an 8 x 3 priority encoder. Assume that the three outputs xyz from the priority encoder are used to provide a vector address of the form 101xyz00. Use the eight vector addresses starting from the one with the highest priority?                                                                                                                                                                                                                                                           | 4 | 3 | 4 | 3   |
| 15. a) | Explain Segmented-Page Mapping with neat block diagram? Explain the format of a typical segment descriptor?                                                                                                                                                                                                                                                                                                                                                                                                           | 4 | 2 | 5 | 3   |
| b)     | Consider a machine with a clock frequency of 3 GHz executing 4000 instructions with an average CPI of 3. Suppose we use a new compiler on the same program for which the new instruction count is 7000, and a new CPI is 1.8, running on a faster machine with a clock rate of 4 GHz. What will be the speedup achieved?                                                                                                                                                                                              | 4 | 3 | 5 | 3   |
| 16. a) | Explain typical one stage Arithmetic Logic Shift Unit (ALU) with block diagram and function table?                                                                                                                                                                                                                                                                                                                                                                                                                    | 4 | 2 | 1 | 2   |
| b)     | What is interrupt? Explain interrupt cycle with neat flowchart?                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4 | 2 | 2 | 1,2 |
| 17.    | Answer any <i>two</i> of the following:                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4 | 3 | 3 | 3   |
| a)     | Perform the multiplication of (+9) and (-13) using Booth's multiplication algorithm?                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 | 3 | 4 | 3   |
| b)     | A DMA controller transfers 16 bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at a rate of 2400 characters per second. The CPU is fetching and executing instructions at an average rate of 1 million instructions per second. By how much will the CPU be slowed down because of the DMA transfer?                                                                                                                                                         | 4 | 3 | 5 | 3   |
| c)     | A computer employs RAM chips of 256 x 8 and ROM chips of 1024 x 8. The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units, each with four registers. A memory-mapped I/O configuration is used. The two highest order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers.<br>a. How many RAM and ROM chips are needed?<br>b. Draw a memory-address map for the system.<br>c. Give the address range in hexadecimal for RAM, ROM, and interface | 4 | 3 | 5 | 3   |

M : Marks; L: Bloom's Taxonomy Level; CO; Course Outcome; PO: Programme Outcome

|      |                               |     |
|------|-------------------------------|-----|
| i)   | Blooms Taxonomy Level – 1     | 20% |
| ii)  | Blooms Taxonomy Level – 2     | 40% |
| iii) | Blooms Taxonomy Level – 3 & 4 | 40% |

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